L01 — Introduction

Course overview and digital character pipeline

Study summary generated from lecture slides. ETH AI for Digital Characters.

Course goals

- Build conversational digital characters end-to-end
- Speech recognition, LLM responses, speech synthesis, animation
- Autonomous agents, personality, and emotions

Assessment

- Written exam: 120 minutes, 120 points, no summary sheet
- Non-programmable calculator and neutral dictionary allowed
- Optional quizzes in 7 lectures: 25/35 correct -> +0.25 grade bonus

Pipeline architecture

- User speech -> Speech Recognition -> transcript
- Transcript + emotion/intention/personality -> Chatbot (LLM or dialogue tree)
- Response text -> Speech Synthesis + Animation Synthesis
- Feedback loop: user emotion detected from text, speech, video

Course topics (lecture map)

- Affective Computing -> Deep Learning -> Speech Recognition
- LLMs (3 lectures) -> Speech Synthesis (2) -> Animation (2)
- Autonomous Agents -> Applications -> Reinforcement Learning

Use cases

- Digital Einstein: ETH branding, chatbot + TTS + animation
- Healthcare, education, museums, metaverse companions
- Location-based and mixed-reality experiences

Speech synthesis history (preview)

- Concatenative -> statistical parametric -> neural end-to-end
- Training costs of modern LLMs are substantial (industry context)